
コンピュータビジョン特論A/B 現代脳計算論

Computer Vision A/B Modern Neural Computation

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Basic Image Processing

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□ Topic

- Color Histogram
- Binarization

- Smoothing
- Remove Salt-and-Pepper-Noise
- Edge Extraction
- Local Feature
 - ◆ Corner Detector

Local Feature/Corner Detector

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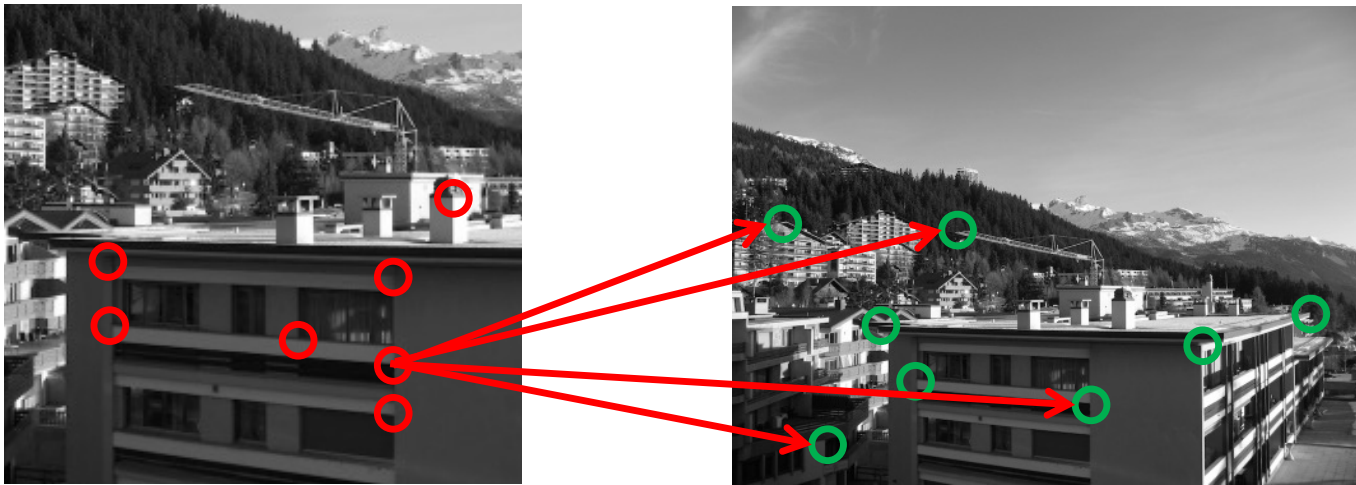
- Local Feature & Feature Detector
 - Corner Detector



Feature Detection

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Local Feature Matching



Which feature point is the corresponded point?

Feature Detection

- Corner region
- Edge region
- Flat region

Hessian Corner Detector

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□ Image Corner

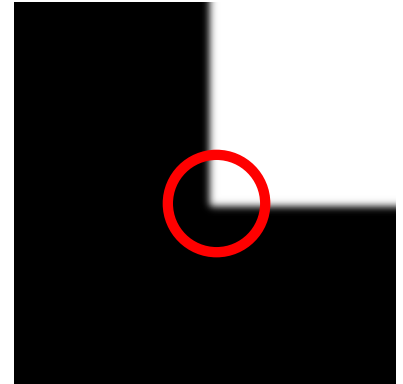
- Change points in the x, y and I coordinate system

□ Image Curvature

- Gauss Curvature: $K = \frac{\det(H)}{(1+I_x^2+I_y^2)^2}$
- Hessian matrix $H = \begin{pmatrix} I_{xx}(x, y) & I_{xy}(x, y) \\ I_{xy}(x, y) & I_{yy}(x, y) \end{pmatrix}$, $\det(H) = I_{xx}I_{yy} - I_{xy}^2$
- Image corner is a high curvature point

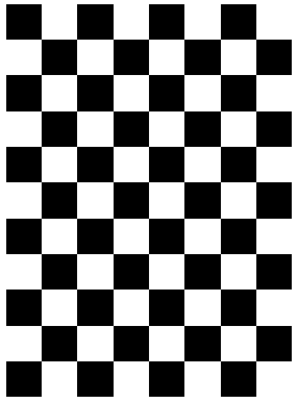
□ Hessian Corner detector

- Find the maximum points (local maximum) for $\det(H)$
 - ◆ $\det(H)$ is a numerator (分子) of the Gauss Curvature K

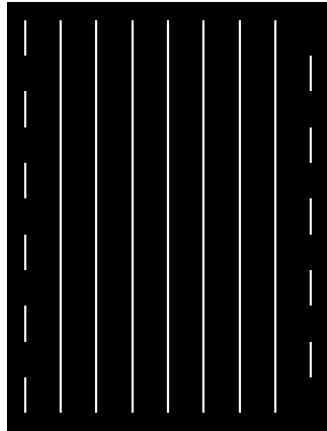


Hessian Corner Detector

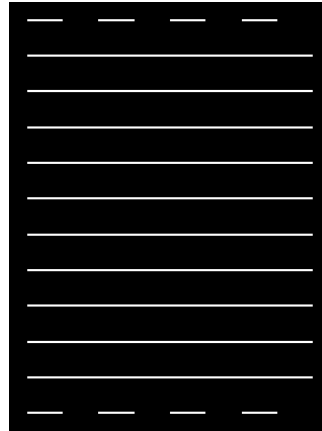
43



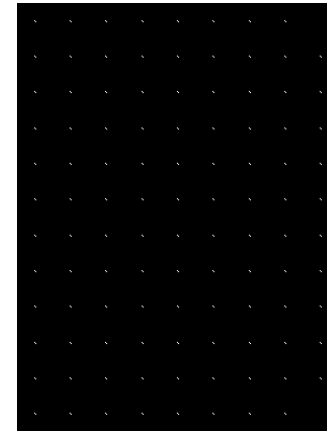
Original



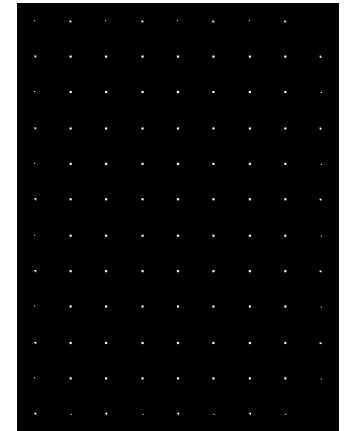
I_{xx}



I_{yy}



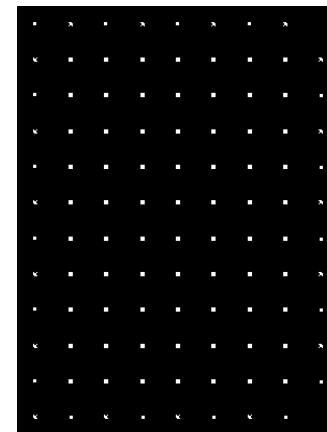
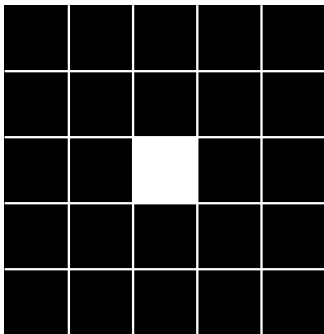
I_{xy}



$\det(H) = I_{xx}I_{yy} - I_{xy}^2$

□ Find local maximum for the $\det(H)$ image

- Calc. Difference for surrounding pixels



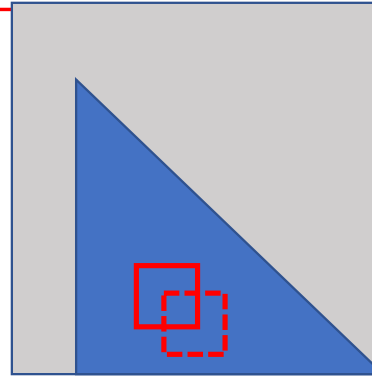
Detected Corner

Moravec Corner Detector

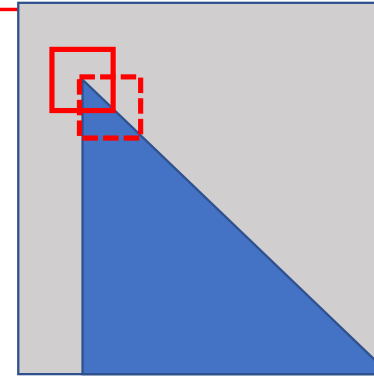
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□ Moravec Corner

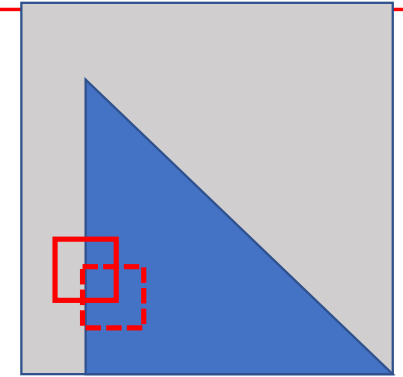
- Similarity based detector
- Calculate similarity with neighboring regions
 - ◆ Flat: high similarity
 - ◆ Corner: low similarity
 - ◆ Edge: high similarity in a specific direction



flat



corner



edge

□ Corner Strength

- Find local maximum of $\min(C_{xx}, C_{yy}, C_{xy})$
- SSD: Sum of Squared Difference
 - ◆ $C_{SSD}(u, v) = \sum_{j=0}^{N-1} \sum_{i=0}^{M-1} (I(u + i, v + j) - I(i, j))^2$
 - $I(u, v)$: image
 - (u, v) : pixel variation

Harris Corner Detector (1/4)

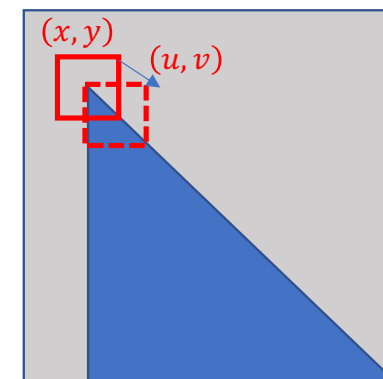
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□ Detect Edge & Corner points (improved method of Moravec corner detector)

- $E(u, v) = \sum_{x,y} w(x, y) (I(x + u, y + v) - I(x, y))^2$
 - ◆ $w(x, y)$: window function (e.g. rectangular window, gaussian function)
 - ◆ u, v : shift

□ Feature computation

- $E(u, v) = \sum_{x,y} w(x, y) (I(x + u, y + v) - I(x, y))^2$
 $\approx \sum_{x,y} w(x, y) (I(x, y) + uI_x + vI_y - I(x, y))^2$
 - ◆ $f(x + u, y + v) \approx f(x, y) + uf_x(x, y) + vf_y(x, y)$: Taylor expansion
$$= \sum_{x,y} w(x, y) \left((I_x \ I_y) \begin{pmatrix} u \\ v \end{pmatrix} \right)^2 = \sum_{x,y} w(x, y) \left((I_x \ I_y) \begin{pmatrix} u \\ v \end{pmatrix} \right)^T \left((I_x \ I_y) \begin{pmatrix} u \\ v \end{pmatrix} \right)$$
$$= \sum_{x,y} w(x, y) (u \ v) \begin{pmatrix} I_x \\ I_y \end{pmatrix} (I_x \ I_y) \begin{pmatrix} u \\ v \end{pmatrix} = (u \ v) \left(\sum_{x,y} w(x, y) \begin{pmatrix} I_x \\ I_y \end{pmatrix} (I_x \ I_y) \right) \begin{pmatrix} u \\ v \end{pmatrix}$$
$$E(u, v) \approx (u \ v) M \begin{pmatrix} u \\ v \end{pmatrix}, M = \sum_{x,y} w(x, y) \begin{pmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{pmatrix}$$



Harris Corner Detector (2/4)

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□ Feature computation

- $E(u, v) \approx (u \ v)M \begin{pmatrix} u \\ v \end{pmatrix}$, $M = \sum_{x,y} w(x, y) \begin{pmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{pmatrix}$

□ If $w(x, y) = 1$

- $M = \sum_{x,y} \begin{pmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{pmatrix}$

- (a): corner

- ◆ $\sum_{x,y} I_x^2 = \lambda_x > 0, \sum_{x,y} I_y^2 = \lambda_y > 0$

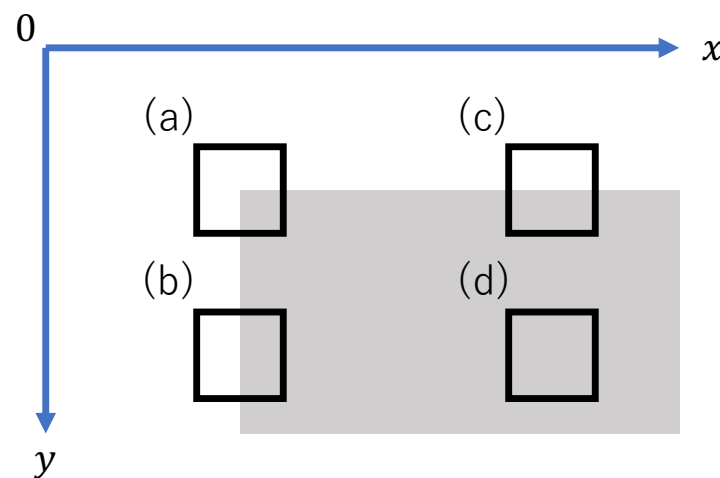
- ◆ $M = \begin{pmatrix} \lambda_x & 0 \\ 0 & \lambda_y \end{pmatrix}$

- (b), (c): edge, I_x or $I_y = 0$

- ◆ $M = \begin{pmatrix} \lambda_x & 0 \\ 0 & 0 \end{pmatrix}$ or $\begin{pmatrix} 0 & 0 \\ 0 & \lambda_y \end{pmatrix}$

- (d): flat, I_x and $I_y = 0$

- ◆ $M = \mathbf{0}$



General Case (Mを対角化)

$$M = U \begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix} U^{-1}$$

α, β : eigen values

U : rotation matrix

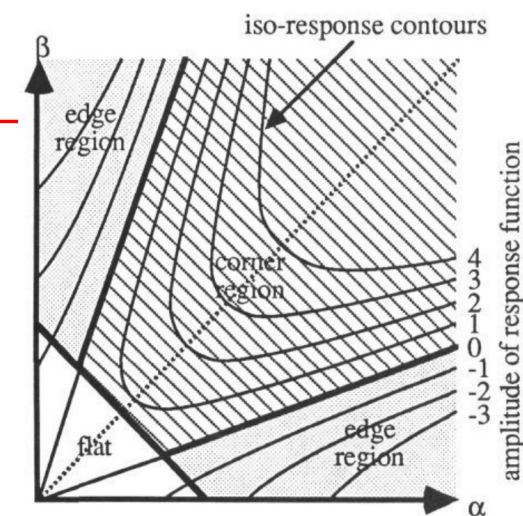
Harris Corner Detector (3/4)

Feature computation

- $E(u, v) \approx (u \ v)M \begin{pmatrix} u \\ v \end{pmatrix}$, $M = \sum_{x,y} w(x, y) \begin{pmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{pmatrix}$
- Computing eigen value is high time cost

Harris Corner Detector

- Detect Corner without computing eigen values
- Cornerness $C = \det(M) - k(\text{tr}(M))^2$
 - ◆ k : constant parameter (0.04 ~ 0.06)
 - ◆ $\det(M) = \alpha\beta$, $\text{tr}(M) = \alpha + \beta$



[C. Harris, M. Stephens, 1998]

General Case (Mを対角化)

$$M = U \begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix} U^{-1}$$

α, β : eigen values

U : rotation matrix

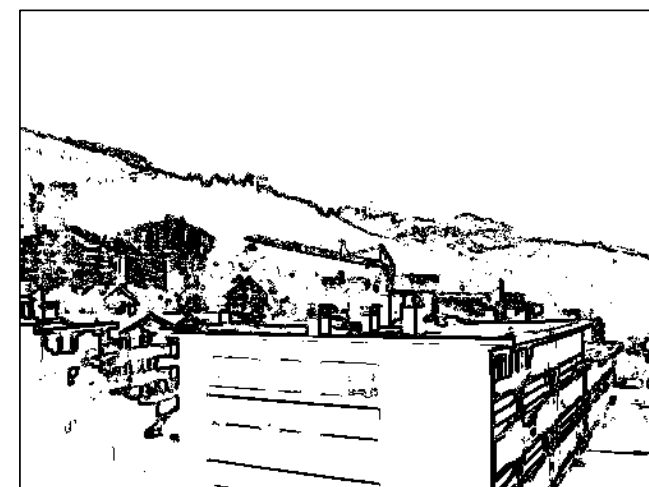
Harris Corner Detector (4/4)



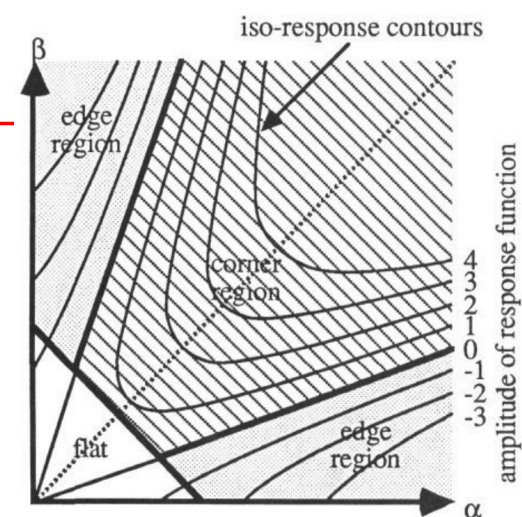
$C \gg 0$ (Corner)



$C \ll 0$ (Edge)



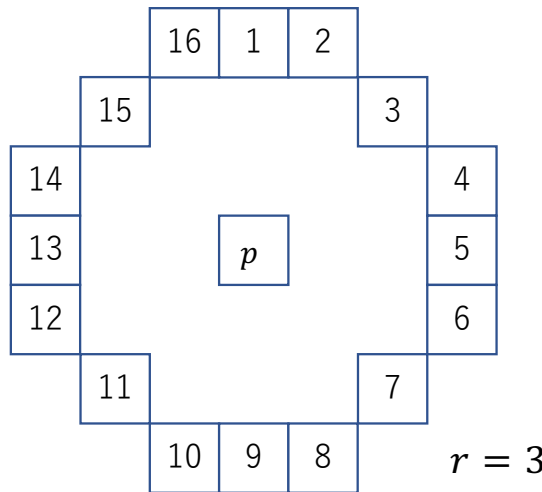
$|C| < \varepsilon$ (Flat)



[C. Harris, M. Stephens, 1998]

Segment Test

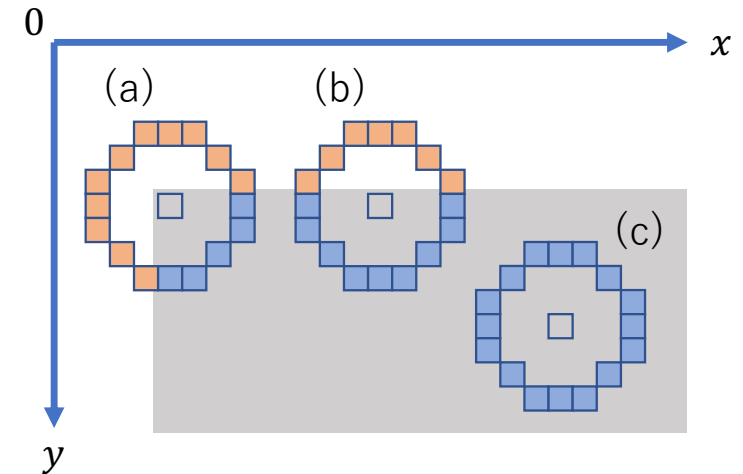
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If $|I_p - I_i| < \Delta I$,

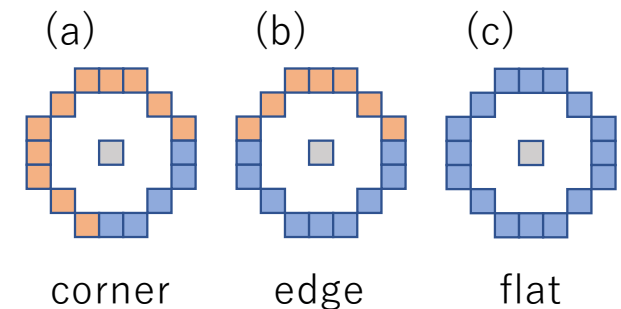


Otherwise,



How to find a corner

1. Check the 1st pixel, and the 9th pixel
 - ◆ If both pixels are , the pixel p is not corner.
2. Check the 5th pixel, and the 13th pixel like 1.
3. Check other pixels
 - ◆ If 12 contiguous pixels are , the pixel p is corner



FAST Corner Detector (1/2)

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□ Segment Test

- Find contiguous pixels
- Search order is not optimal

□ FAST (Features from Accelerated Segment Test)

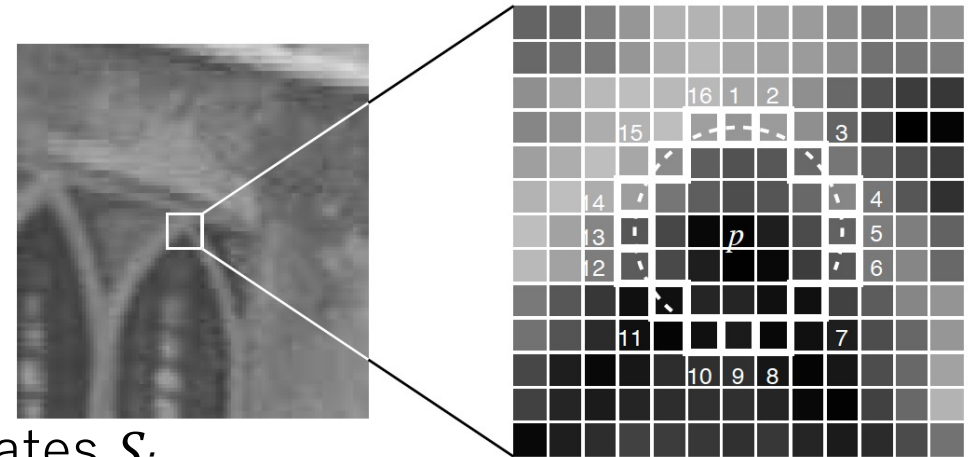
- Labeling surrounding pixels as three states S_i

$$\blacklozenge S_i = \begin{cases} d, & I_i \leq I_p - \Delta I \quad (\text{darker}) \\ s, & |I_p - I_i| < \Delta I \quad (\text{similar}) \\ b, & I_p + \Delta I \leq I_i \quad (\text{brighter}) \end{cases}$$

- Labeled states are represented as feature vector P

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
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- Distinguish whether the pixel p is a corner using a decision tree



[E. Rosten, T. Drummond, 2006]

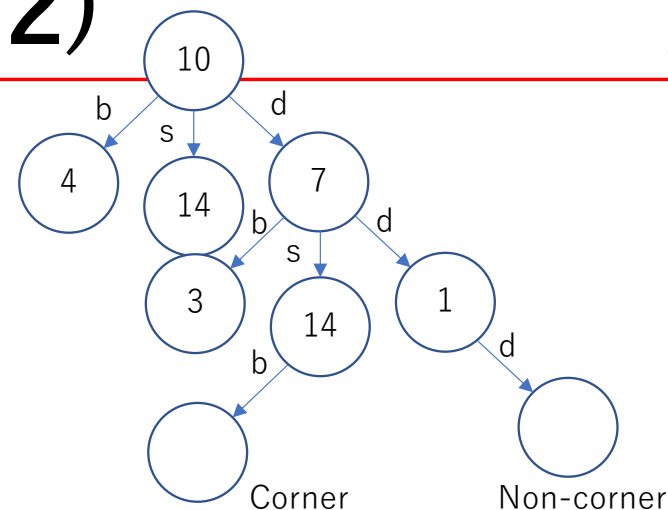
FAST Corner Detector (2/2)

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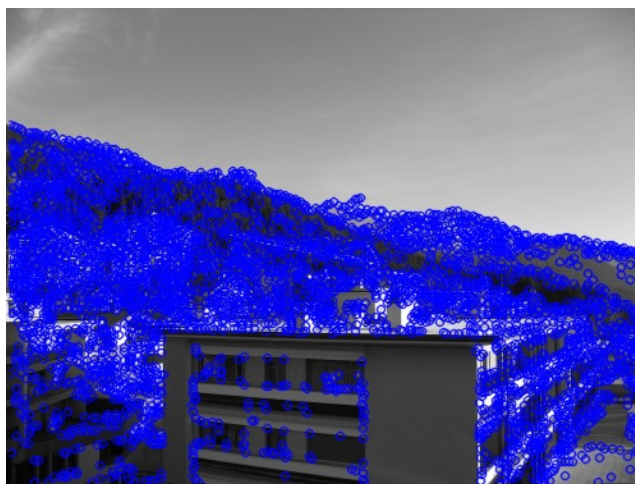
Decision Tree

- Distinguish whether the pixel p is a corner
- Tree is trained using machine learning

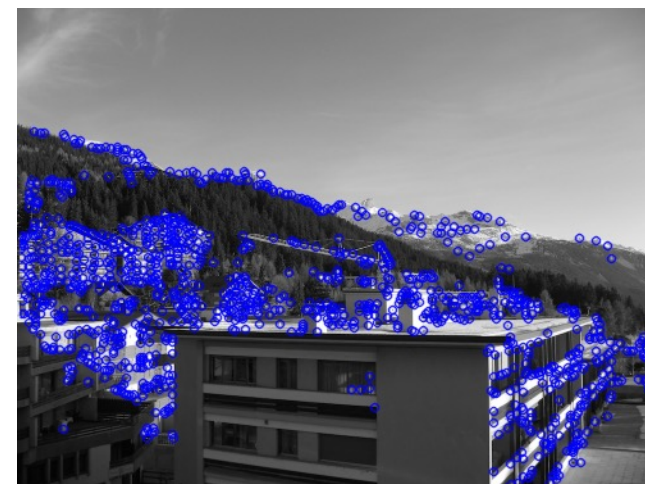
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----



Original Image



$\Delta I=10$ (many non-corner points are included)

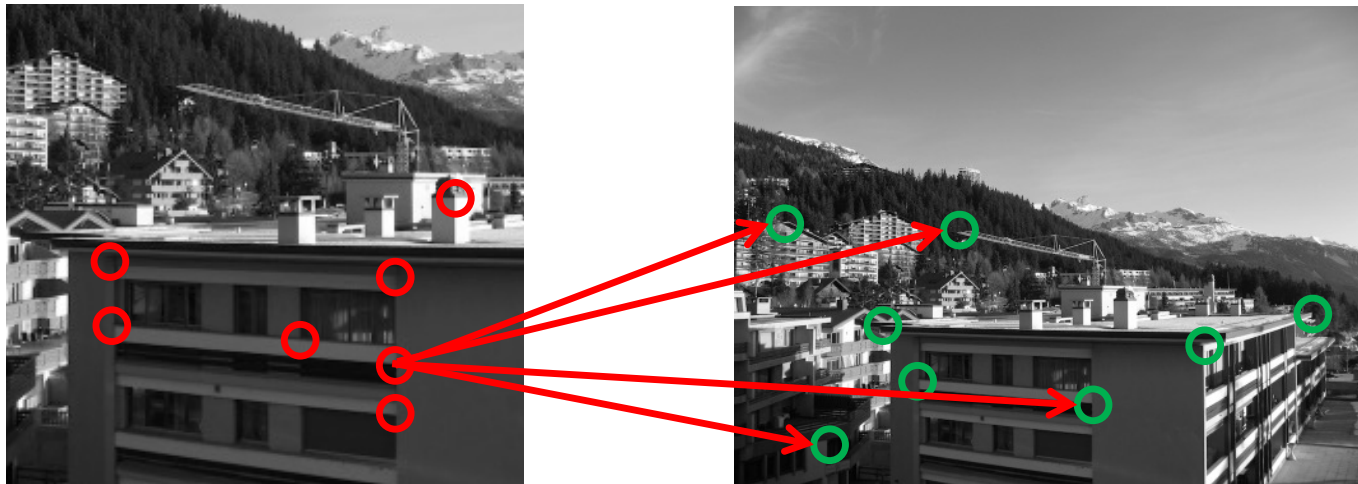


$\Delta I=50$

○ Feature Detection

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▣ Local Feature Matching

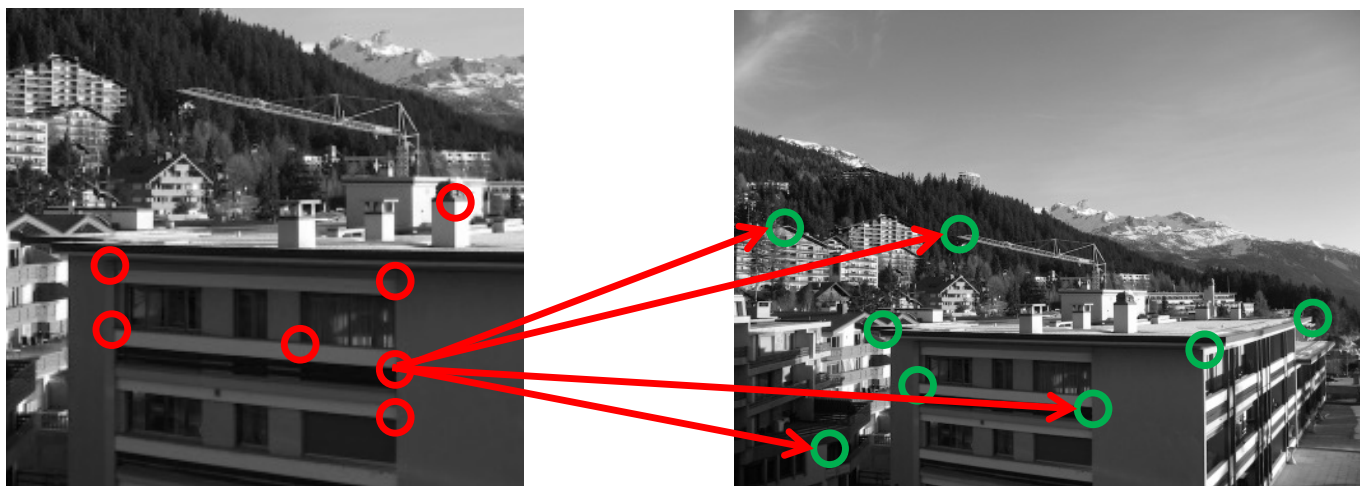


Which feature point is the corresponded point?

Feature Detection

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Local Feature Matching



Which feature point is the corresponded point?

Feature Detection

- Corner region
- Edge region
- Flat region

